Integer solutions in feasible regions

1

a $(6,12),(7,12),(8,12),(8,11),(8,10),(7,11)$

b $(0,2),(1,2),(2,2),(0,1),(1,1),(0,0),(1,0)$

c $(5,6),(5,5),(5,4)$

d $(-6,-7),(-6,-6),(-6,-5),(-5,-5),(-5,-4),(-5,-3)$

e $(6,7),(5,6),(6,6),(7,6),(5,5),(6,5),(7,5),(6,4)$

2

a

i $(0,3),(1,3),(0,2),(1,2),(0,1),(0,0)$ ii $(0,3)$

b

i $(5,7),(5,6),(5,5),(5,4),(6,5)$ ii $(5,4)$

c

i $(6,7),(5,6),(6,6),(7,6),(5,5),(6,5),(7,5),(6,4)$

ii $(5,6)$

3

a

i $(2,4),(3,4),(4,4),(5,4),(6,4),(2,3),(3,3)$

ii $(2,3)$

b

i $(-6,-4),(-5,-4),(-6,-5),(-5,-5),(-4,-6)$

ii $(-5,-4)$

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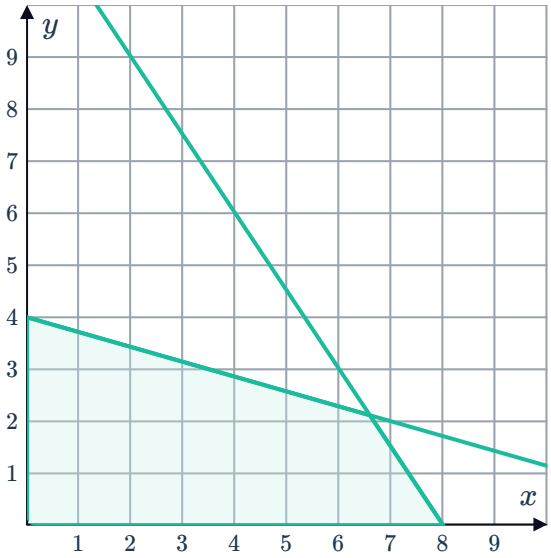
Integer solutions in linear programming applications

4

a A number of students must be a whole number as it is not possible to have a part of a student.

b $x \geq 0, y \geq 0, 3x + 2y \leq 24$ and $x + 3.5y \leq 14$

c



d $P = 200x + 220y$

e

Possible solutions	Profit (dollars)
$(0, 4)$	880
$(7, 0)$	1400
$(6, 2)$	1640

f \$1640

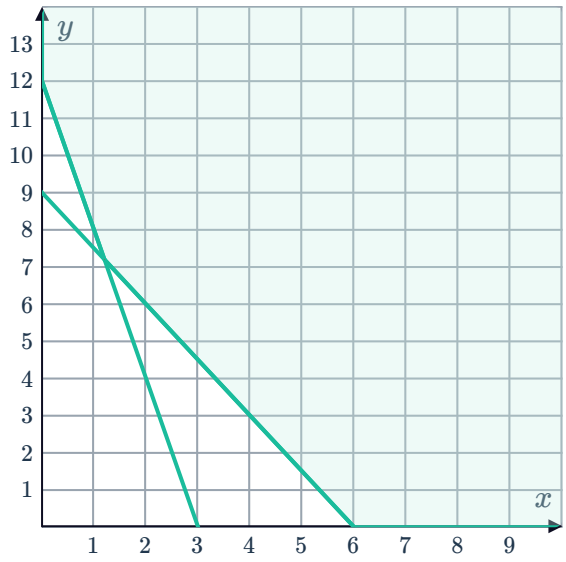
g 6 English students and 2 Mathematics students

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5

a $x \geq 0, y \geq 0, 8x + 2y \geq 24$ and $6x + 4y \geq 36$

b



c $C = 25.8x + 15.2y$

d $(1, 8), (2, 6), (0, 12), (6, 0)$

e

Possible solutions	Ingredients cost ($)
$(0, 12)$	182.40
$(6, 0)$	154.80
$(1, 8)$	147.40
$(2, 6)$	142.80

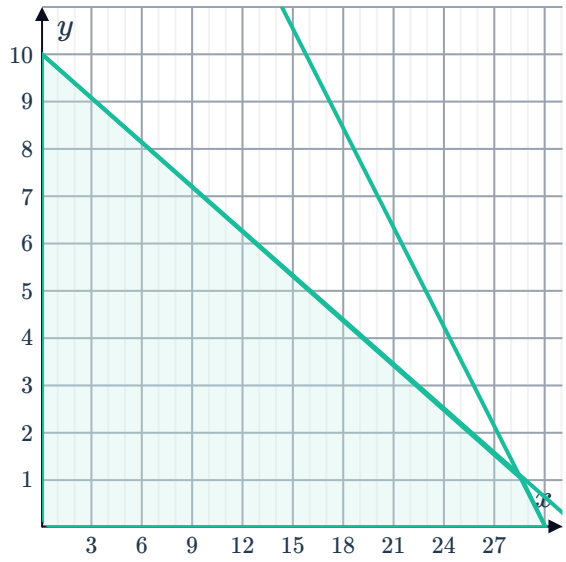
f $142

g 2 cakes and 6 muffins

6

a $x \geq 0, y \geq 0, 175x + 250y \leq 5250$ and $0.25x + 0.8y \leq 8$

b



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c $P = x + y$

d $(0, 10), (25, 2), (28, 1), (30, 0)$

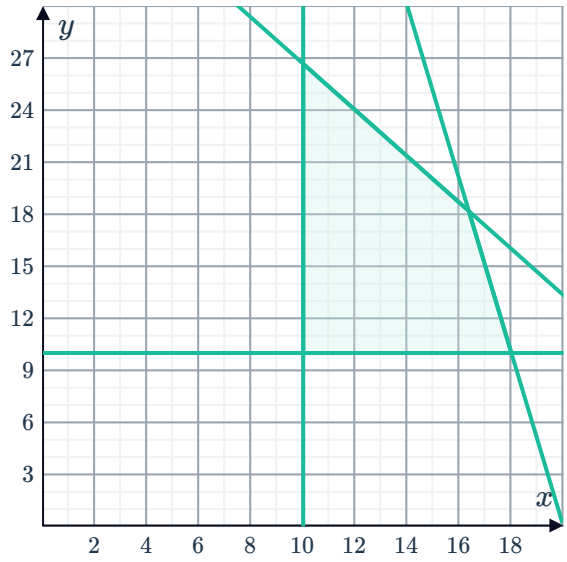
e 30 plants

f 30 carrots and 0 parsnips

7

a $x \geq 10, y \geq 10, 50x + 10y \leq 1000$ and $20x + 15y \leq 600$

b



c $V = 5x + 2y$

d $(16, 18), (10, 26), (18, 10)$

e

Possible solutions	Value ($)
$(16, 18)$	119
$(0, 26)$	102
$(18, 0)$	110

f $116

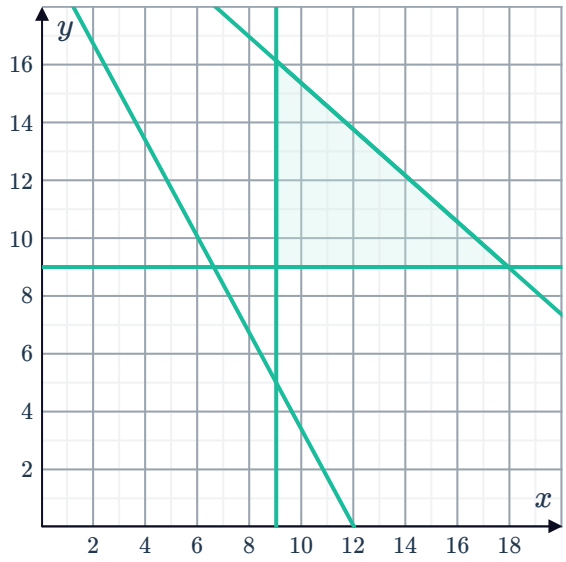
g 16 tomato plants and 18 strawberry plants

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8

a $x \geq 9, y \geq 9, 25x + 15y \geq 300$ and $12x + 15y \leq 350$

b



c $V = 15x + 12y$

d $(18, 9), (9, 16), (9, 9)$

e

Possible solutions	Value ($)
$(18, 9)$	378
$(9, 16)$	327
$(9, 9)$	243

f $378

g 18 protein powder sachets and 9 packets of pancake mix

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